Automatic Itinerary Reconstruction from Texts

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Objectives

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Traverser Champagny-le-Haut et contourner le hameau de Friburge. Vous apercevrez le Lac de la Plagne puis marcher jusqu'au refuge au sud du lac de Grattaleu.

Objectives

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Traverser Champagny-le-Haut et contourner le hameau de Friburge. Vous apercevrez le Lac de la Plagne puis marcher jusqu'au refuge au sud du lac de Grattaleu.

Who understand ?

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Cross Champagny-le-Haut and get around from the north of hamlet Friburge. You will see the Lac de la Plagne then walk to the refuge south of lake Grattaleu.

better ? Where is it ?

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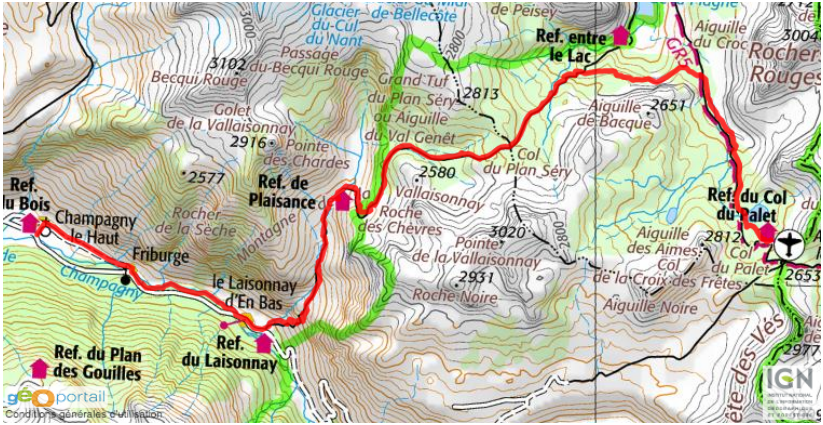

and now ?

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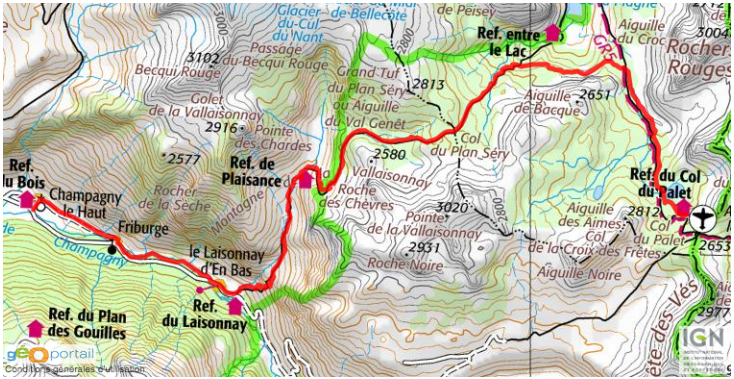
From text...

Traverser Champagny-le-Haut et contourner le hameau de Friburge. Vous apercevrez le Lac de la Plagne puis marcher jusqu'au refuge au sud du lac de Grattaleu.

...to map.



Objectives

Automatic itinerary reconstruction from texts

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Main problems

1. Expression of space and motion in language
2. Automatic information extraction
3. Assigning location to spatial information
4. Route calculation

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Main problems

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1 **Expression of space and motion in language** (Talmy,1985)

- spatial named entities (toponyms)
- geographical terms
- spatial relation
- spatial prepositions, spatial adverb, etc
- verbs of visual perception
- verbs of displacement, motion or location

Objectives

Main problems

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2 Automatic information extraction

- Named Entity Recognition (NER)
  - learning techniques, ad-hoc rules, etc
  - hybrid systems based on **syntactico-semantic patterns**
  - types of named entities : date, person, **spatial**, etc
- Natural Language Processing (NLP)
  - Part-of-speech (POS) tagging

Objectives

Main problems

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3 Assigning location to spatial information

- Toponym resolution (Geocoding)
  - use of geographical resources

- Toponym disambiguation (Smith and Mann, 2003)
  - **Referent ambiguity** : the same name used for several places
  - **Reference ambiguity** : several names for the same place
  - **Referent class ambiguity** : geo / non-geo ambiguity

  - Unreferenced toponyms ambiguity

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4 Route calculation

- Route planning
- Spatial analysis
  - using topological, geometric, or geographic properties
- Spatio-temporal analysis

Corpus

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Corpus of experiments

- 1295 French hike descriptions
- automatically collected from the Web
- collected with **GPS track**

Body of reference

- 30 French hike descriptions **manually annotated**
- controlled manual tagging tool under development

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Geoparsing

Automatic annotation of spatial expressions

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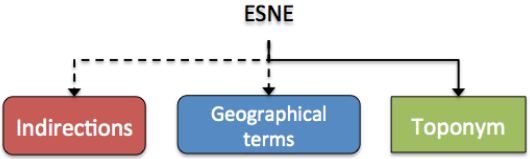
Expanded spatial named entities (ESNE)



- Spatial named entities (mandatory)
  - Toponym + sub-type (feature types)
  - e.g. Vienna, Danube river, lake Grattaleu, etc
- Geographical terms (optional)
  - e.g. refuge, bridge, hamlet, etc
- Indirections (optional)
  - e.g. the north of, from the left, etc

1 : left side of the Danube river

2 : the refuge south of lake Grattaleu

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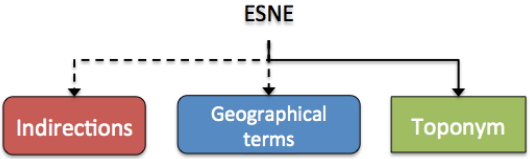

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Expression of motion

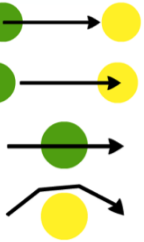
- **motion verbs**
  - to go, to leave, etc
- verbs of visual perception
  - to see, to glimpse, etc
- "topographic" verbs
  - to converge, to overhang, etc
- location verbs
  - to locate, to be, etc

Polarity of motion verbs

Initial verbs :

Final verbs :

Median verbs :



VTo structures

- relationship between toponyms or ESNE and expressions of motion

1 : **Reach** the left side of the Danube river

2 : **Walk** to the refuge south of lake Grattaleu

Geoparsing

Automatic annotation of spatial expressions

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Combine language and spatial analysis to order places and reconstruct the path

- Use information extracted from textual description
  - places not reached (perception, negation,...)
  - indirection (direction, distance,...)
  - motion expressions, polarity of the displacement (find origin and destination)
- Use spatial analysis to link places

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Minimum weight spanning tree

- Geoparsing
  - polarity of displacement
  - places not reached
- Geocoding
  - distances

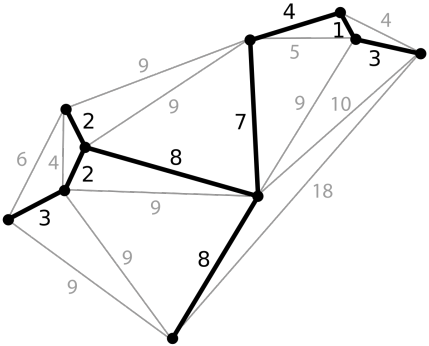

FIGURE : Minimum weight spanning tree (source : wikipedia)

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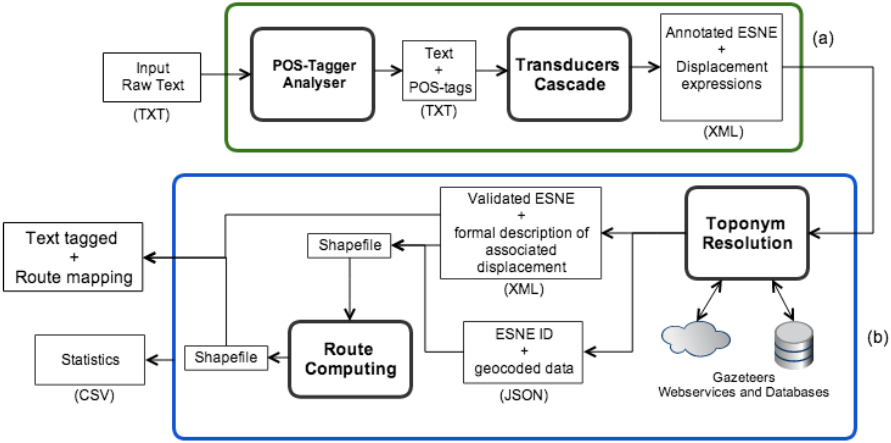

FIGURE : Block diagram of our processing chain

Geoparsing

Finite-state transducers cascade

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Our linguistic rules are implemented using a finite-state transducers cascade

Input

- Raw text + POS tags

example : {Walk,.V} {to,.PREP} {the,.ART} {refuge,.N} {south,.N} {of,.PREP} {lake,.N} {Grattaleu,.NPr} {.,.PUN}

Main transducers

1. Indirections (spatial relations)
2. Candidate toponyms (+ sub-type)
3. ESNE
4. Motion, perception (*VTo Structures*)

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Example of transducer

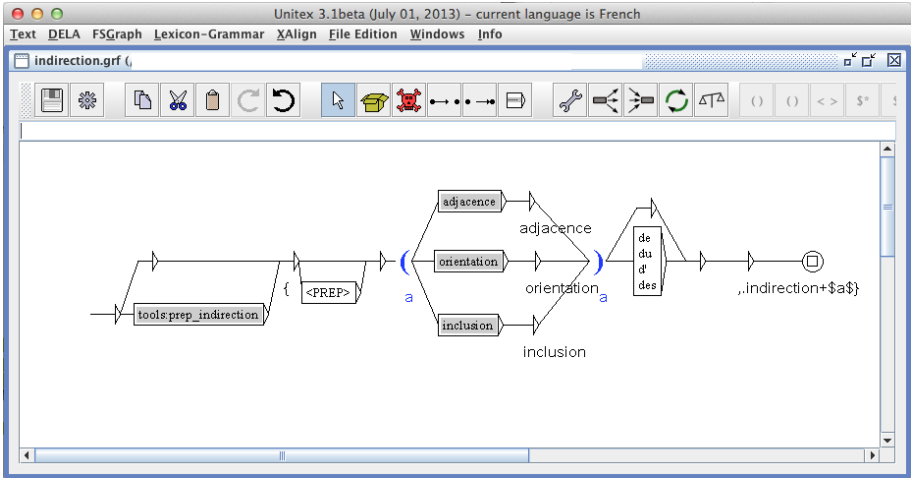

FIGURE : Example of transducer in the Unitex platform

Result of geoparsing

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Cross Champagny-le-Haut and get around from the north of hamlet Friburge. You will see the Lac de la Plagne then walk to the refuge south of lake Grattaleu.

Execution of the transducers cascade

1. Indirections (spatial relations)
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↓

- Cross Champagny-le-Haut
- get around from the north of hamlet Friburge
- see the Lac de la Plagne
- walk to the refuge south of lake Grattaleu

Geoparsing

Finite-state transducers cascade

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Preprocessing	(a)	(b)	(c)	Precision	Recall
Automatic POS	583	581	595	**90.02%**	**94.68%**
POS manually corrected	583	559	636	98.29%	99.42%

TABLE : Evaluation of our geoparsing process

(a) number of toponyms manually annotated

(b) number of relevant toponyms automatically annotated

(c) number of toponyms automatically annotated

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Geocoding

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Querying of gazetteers

- **BDNyme** : French toponymic database provided by IGN
- **Geonames** : geographical database (geocoding Web Services)

Toponyms ambiguities

- **50%** of toponyms found in gazetteers have **more than 1 result**
- **30%** of toponyms **are not found** in gazetteers (problem of fine-grain toponyms)

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Geocoding

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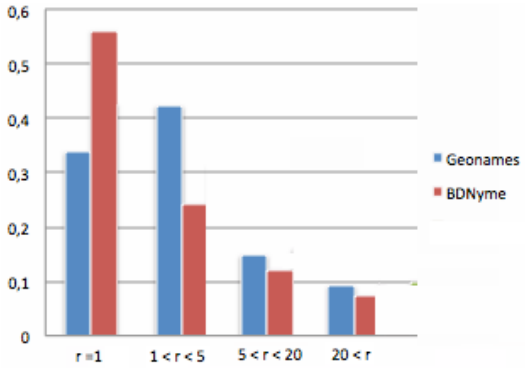
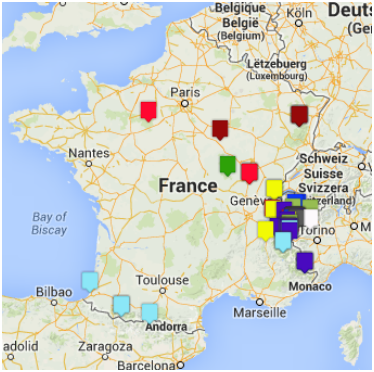
Referent ambiguity : same name used for several places





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Structural ambiguity : ambiguity on words constituting the name

- 46% of toponyms are associated with feature types

FR	EN	Frequency
col	pass	20
village	town	20
hameau	hamlet	20
rue	road	17
chemin	path	15
chalet	cottage	13
refuge	refuge	11
pont	bridge	11
lac	lake	8
chapelle	chapel	8

TABLE : Most frequent terms associated with toponyms

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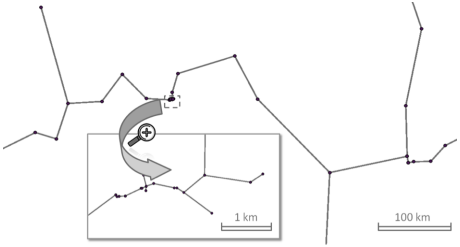
Toponyms desambiguation



Result of geocoding + real GPS track

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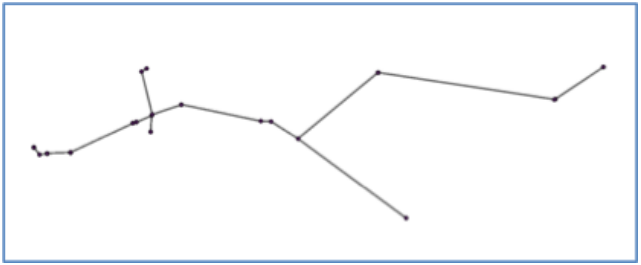
Minimum spanning tree (euclidian distances)



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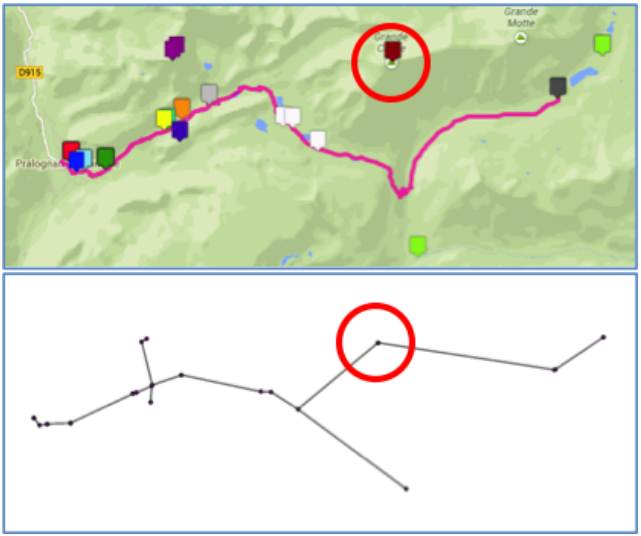

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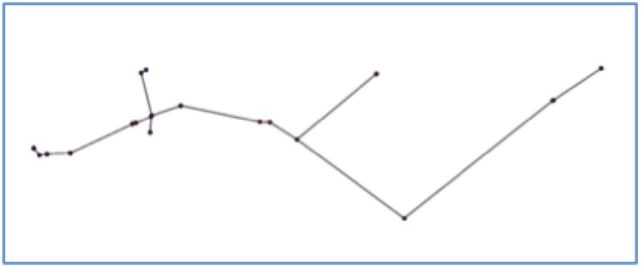
Minimum spanning tree using information extracted from the text



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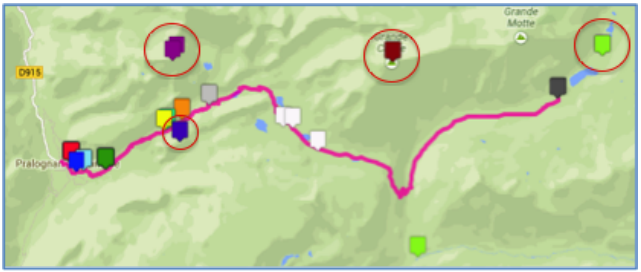

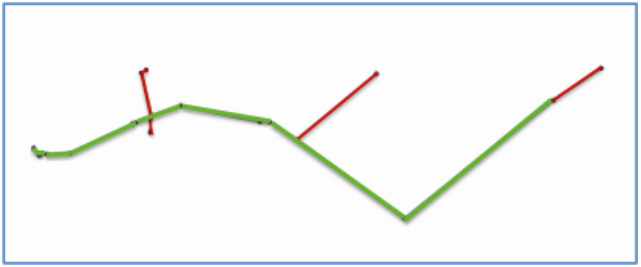

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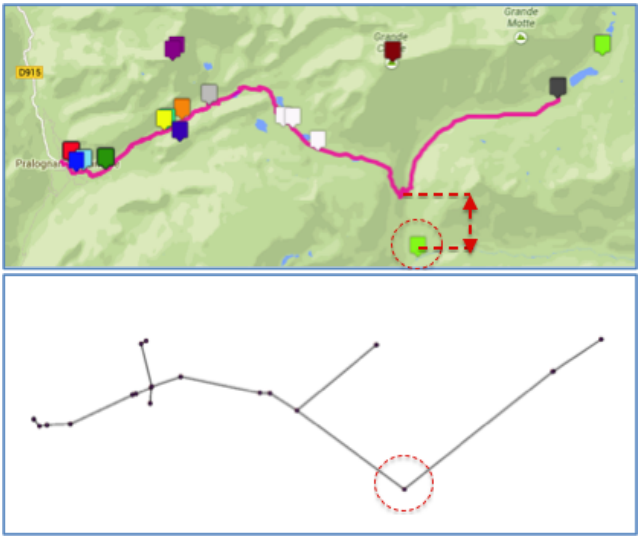

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Minimum spanning tree using information extracted from the text

FIGURE : Automatic itinerary VS real GPS trace

Online geoparsing and geocoding tool

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http ://erig.univ-pau.fr/PERDIDO/

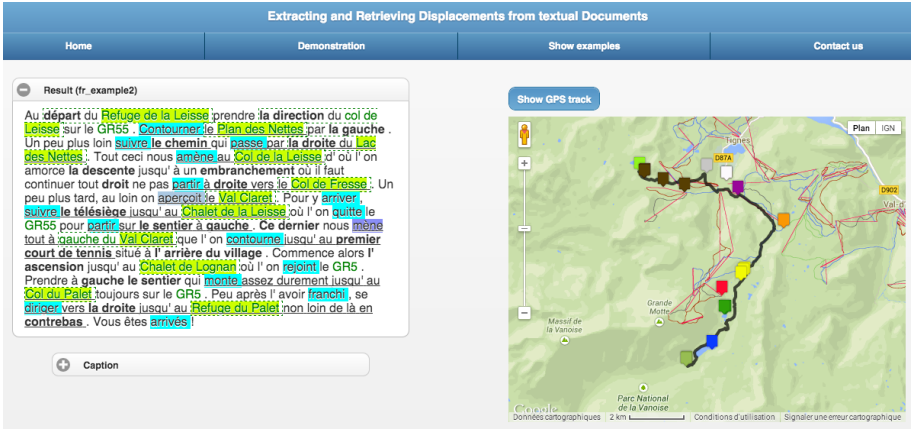

FIGURE : Screenshot of the online geoparsing and geocoding tool

OUTLINE

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Expanded geoparsing process

- spatial context of toponyms is extracted : feature types, displacement, perception, etc

Geocoding and itinerary reconstruction

- minimum spanning tree weighted using information (places not reached, etc)
- information extracted from the text allows a better interpretation

Mixing spatial and textual analysis

- first results are encouraging

Work in progress

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Multilingual

- French, Spanish, Italian

Toponyms desambiguation

- Clustering algorithm : DBSCAN (Density-Based Spatial Clustering of Applications with Noise)
- Geocoding of unreferenced toponyms

Full paper accepted in ACM SIGSPATIAL 2014

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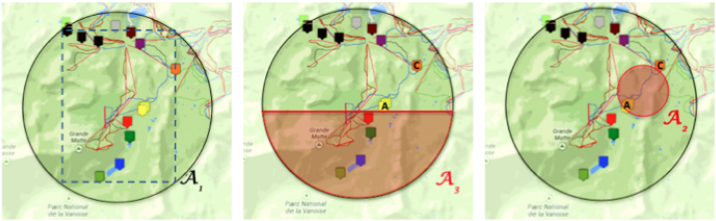

(a) (b) (c)

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Outlook

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Improve the route calculation using other information

- level of the trail
- means of transport (walk, bike, etc)
- similarity, complementarity and proximity
- temporal information

Outlook

The use of temporal information

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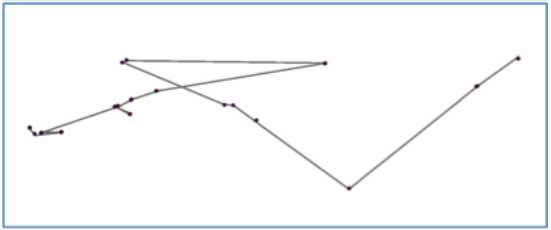
Preliminary experiments

- Ordering toponyms as thez occur in the text
- The need of temporal context



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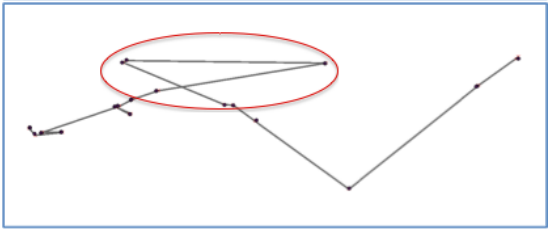

Thank you for your attention

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